

A Complete Fourier Spectral Atlas for Constant- f Rotating Shallow Water

HCS Research Program

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Abstract

We give a global same-clock theorem for the linear rotating shallow-water equations on the doubly periodic 2π -torus. Each lattice mode is a three-by-three skew-Hermitian block. Its exact zero, plus, and minus projectors resolve the stationary geostrophic/potential-vorticity branch and the two inertia-gravity branches, including all zero-parameter faces. We also prove the sum-of-two-squares shell multiplicities, a necessary and sufficient periodicity test for finite Fourier support, and the global unitary but noncompact/non-Schatten boundary. A deterministic ledger and independent symbolic and numerical paths audit the claims. The model has constant f ; no beta-plane Rossby assertion is made.

1 Model and statement

Let $u = (u_1, u_2)$ and ϕ be complex fields on $\mathbb{T}^2$, and let

$$u_t + fJu + c\nabla\phi = 0, \quad \phi_t + c\nabla \cdot u = 0, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Here $f \in \mathbb{R}$, $c \geq 0$, and t is physical time. With $e_n(x) = e^{in \cdot x}$, $n \in \mathbb{Z}^2$, the coefficient vector $X_n = (\hat{u}_{1,n}, \hat{u}_{2,n}, \hat{\phi}_n)^T$ obeys

$$\dot{X}_n = G_n X_n, \quad G_n = \begin{pmatrix} 0 & f & -icn_1 \\ -f & 0 & -icn_2 \\ -icn_1 & -icn_2 & 0 \end{pmatrix}, \quad \omega_n = (f^2 + c^2|n|^2)^{1/2}.$$

For $c > 0$, the Fourier multiplier is taken on its maximal graph domain

$$D(G) = \left\{ X \in L^2 : \sum_{n \in \mathbb{Z}^2} \|G_n X_n\|_{\mathbb{C}^3}^2 < \infty \right\};$$

trigonometric polynomials form a core and H^1 is contained in this domain. The skew-Hermitian blocks close to a skew-adjoint generator and hence to the global unitary L^2 group. When $c = 0$, the bounded pointwise rotation is treated separately.

Theorem 1 (projector, shell, and operator atlas). *For every n with $\omega_n > 0$, $G_n^* = -G_n$,*

$$G_n^3 + \omega_n^2 G_n = 0, \quad P_0 = I + \frac{G_n^2}{\omega_n^2}, \quad P_{\pm} = \frac{1}{2} \left(-\frac{G_n^2}{\omega_n^2} \mp \frac{iG_n}{\omega_n} \right),$$

and P_0, P_+, P_- are mutually orthogonal projectors with ranks 1, 1, 1. Moreover

$$e^{tG_n} = P_0 + \cos(\omega_n t)(I - P_0) + \frac{\sin(\omega_n t)}{\omega_n} G_n.$$

The P_0 branch is stationary and is the linear potential-vorticity/geostrophic branch; $P_{\pm}$ are the inertia-gravity branches. At $n = 0$, $f \neq 0$ gives $0, \pm i|f|$, while $f = 0$ gives a zero block. For $c > 0$, the Fourier potential-vorticity quantity

$$\hat{\zeta}_n - \frac{f}{c} \hat{\phi}_n, \quad \hat{\zeta}_n = i(n_1 \hat{u}_{2,n} - n_2 \hat{u}_{1,n}),$$

is constant in time.

For $q > 0$, the shell $|n|^2 = q$ has $r_2(q) = 4(d_1(q) - d_3(q))$ modes. A finite Fourier-support solution is T -periodic if and only if $\omega_n T \in 2\pi\mathbb{Z}$ for every participating nonzero branch. The full group on $L^2(\mathbb{T}^2; \mathbb{C}^3)$ is unitary, but for every t , including $t = 0$, it is neither compact nor in any Schatten class.

Proof. Fourier transformation gives G_n . Direct multiplication yields the minimal polynomial $z(z^2 + \omega_n^2)$, and inspection of the entries gives $G_n^* = -G_n$. The three Lagrange polynomials for the roots $0, \pm i\omega_n$ are precisely P_0, P_+, P_- , proving orthogonality and the exponential after reducing its power series modulo the cubic. Taking $\zeta = \partial_1 u_2 - \partial_2 u_1$ gives $\zeta_t = -f \nabla \cdot u$, while $\phi_t = -c \nabla \cdot u$, proving the potential-vorticity identity for $c > 0$. The shell formula is the classical sum-of-two-squares identity. Applying the projectors to a finite superposition proves the periodicity equivalence. Finally, skew-adjointness gives unitarity; an orthonormal sequence of distinct Fourier modes is carried to another orthonormal sequence of norm one, proving noncompactness and the failure of every Schatten condition. $\square$

2 Degeneracies and finite controls

The faces $f = 0$, $c = 0$, and $n = 0$ are not silently folded into the generic projector formula. In particular, $f = c = 0$ is a three-dimensional zero block, whereas $c = 0$ leaves a scalar stationary component at every mode. The ledger contains eight rate cases, including a negative- f retrograde sentinel, and all integer modes in $[-3, 3]^2$. It also checks the shell identity for $q \leq 18$. The periodicity assertion is only for finite support; an arbitrary L^2 state need not be periodic.

3 Independent audit and scope boundary

The producer records 392 blocks, exact integer shell counts, projector residuals, two matrix-exponential times, and potential-vorticity residuals. The checker rebuilds every block independently, tests the cubic, all projector products, unitarity, exponential agreement, and every boundary face. SymPy supplies a separate symbolic path; byte replay is exact and all repaired and stale hash mutations are rejected.

The result is source-native Fourier analysis. Its Route-A tuple is

$$(AO_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),$$

with `overall=ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`.

The registered scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. Spatial Fourier shells are not claimed to be arithmetic primitive orbits. No target prime, zero, local factor, root number, automorphy statement, functional equation, or Hilbert–Polya operator is introduced.

Source note

The f -plane doubly periodic normal-mode context is discussed by Mohebalhojeh and Dritschel [1]; Salmon [2] provides Hamiltonian projection context. Beta-plane results are intentionally outside the frozen model.

References

- [1] A. R. Mohebalhojeh and D. G. Dritschel, *Hierarchies of Balance Conditions for the f-Plane Shallow-Water Equations*, *Journal of the Atmospheric Sciences* 58 (2001), 2411–2426. DOI: 10.1175/1520-0469(2001)058<2411:HOBCT>2.0.CO;2.
- [2] R. Salmon, *Semigeostrophic theory as a Dirac-bracket projection*, *Journal of Fluid Mechanics* 196 (1988), 345–358. DOI: 10.1017/S0022112088002733.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic or operator claim. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C217. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked the displayed claims and metadata. This is not external peer review.